WORKSHEET 4.1 · Intersection of 2 Lines

3a) $\frac{x-3}{10} = \frac{y-8}{-4} \Rightarrow \vec{m_1} = (10, -4)$

$\frac{x-33}{-5} = \frac{y+4}{2} \Rightarrow \vec{m_2} = (-5, 2)$

$\vec{m_1} = -2\vec{m_2}$

$\therefore$ ∥.

check points from L_1 $P(3, 8)$ into L_2.

$LS = \frac{3-33}{-5} = \frac{-30}{-5} = 6$

$RS = \frac{8+4}{2} = \frac{12}{2} = 6$

$\therefore LS = RS$

$\therefore$ coincident Lines

b) $x = 6 - 18s$, $y = 12 + 3s$ $\quad \vec{m_1} = (-18, 3)$

$x = 8 - 6t$, $y = 4 + 9t$ $\quad \vec{m_2} = (-6, 9)$

since $\vec{m_1} \not\parallel \vec{m_2}$

$\therefore$ neither

c) $x = 8 + 12s$
$y = 4 - 4s$
$z = 3 - 6s$

$\vec{m_1} = (12, -4, -6)$

$x = 2 - 4t$
$y = 2 + t$
$z = 6 + 2t$

$\vec{m_2} = (-4, 1, 2)$

$\vec{m_1} \not\parallel \vec{m_2}$

$\therefore$ neither

d) $\frac{x+4}{3} = \frac{y-12}{4} = \frac{z-3}{6}$ $\quad \vec{m_1} = (3, 4, 6)$

$\frac{x}{\frac{1}{2}} = \frac{y-10}{\frac{2}{3}} = z + 5$. $\quad \vec{m_2} = (\frac{1}{2}, \frac{2}{3}, 1)$

$\vec{m_1} = 6\vec{m_2}$

$\therefore$ parallel

Pt in L_2: $P(0; 10, -5)$

check in L_1.

$t = \frac{0+4}{3} = \frac{4}{3}$ $\quad t = \frac{10-12}{4} = \frac{-2}{3}$ $\quad t = \frac{-5-3}{6} = \frac{-8}{6} = \frac{-4}{3}$.

since t not unique,

$\therefore$ parallel but distinct

4a) $\vec{r}_1 = (-2,0,-3) + t(5,1,3) \quad t \in \mathbb{R}$

$\vec{r}_2 = (5,8,-6) + u(-1,2,-3) \quad u \in \mathbb{R}$

$\vec{r}_1$: $x = -2+5t$, $y = t$, $z = -3+3t$ ① $t \in \mathbb{R}$

$\vec{r}_2$: $\frac{x-5}{-1} = \frac{y-8}{2} = \frac{z+6}{-3}$ ②

sub ① into ②

$\frac{-2+5t-5}{-1} = \frac{t-8}{2}$ $\qquad$ $\frac{t-8}{2} = \frac{-3+3t+6}{-3}$

$2(-7+5t) = -(t-8)$ $\qquad$ $-3(t-8) = 2(3+3t)$

$-14+10t = -t+8$ $\qquad$ $-3t+24 = 6+6t$

$11t = 22$ $\qquad$ $-9t = -18$

$\boxed{t=2}$ ← SAME! → $\boxed{t=2}$

$\therefore x = -2+5(2) = 8$

$y = (2) = 2$

$z = -3+3(2) = 3$

$\therefore$ P.O.I $(8,2,3)$

b) $x = 1+t$, $y = 1+2t$, $z = 1-3t$ ① $t \in \mathbb{R}$

$x = 3-2u$, $y = 5-4u$, $z = -5+6u$ ② $u \in \mathbb{R}$

$\vec{m}_1 = (1,2,-3)$ $\qquad$ $\vec{m}_2 = (-2,-4,6)$ $\qquad$ $\vec{m}_2 = -2\vec{m}_1$

$\therefore$ parallel.

check pt $(1,1,1)$ into Line ②

$1 = 3-2u$ $\qquad$ $1 = 5-4u$ $\qquad$ $1 = -5+6u$

$1 = u$ $\qquad$ $1 = u$ $\qquad$ $1 = u$

$\therefore$ coincident lines

$\therefore$ infinite solutions

OR// Solve as in part (a) but get

$0t = 0$

infinite solution.

$\therefore (x,y,z) = (1+t, 1+2t, 1-3t)$

c) $\vec{r_1} = (2,-1,0) + t(1,2,-3)$, $t \in \mathbb{R}$ $\quad \vec{m_1} \not\parallel \vec{m_2}$

$\vec{r_2} = (-1,1,2) + u(-2,1,1)$, $u \in \mathbb{R}$

$\vec{r_1}$: $\begin{cases} x = 2+t \\ y = -1+2t \\ z = -3t \end{cases}$ ① $t \in \mathbb{R}$

$\vec{r_2}$: $\dfrac{x+1}{-2} = \dfrac{y-1}{1} = \dfrac{z-2}{1}$ ②

sub ① into ②

$$\frac{2+t+1}{-2} = \frac{-1+2t-1}{1}$$

$$1(3+t) = -2(-2+2t)$$

$$3+t = 4-4t$$

$$5t = 7$$

$$\boxed{t = \frac{7}{5}}$$

$$\frac{-1+2t-1}{1} = \frac{-3t-2}{1}$$

$$-2+2t = -3t-2$$

$$5t = 0$$

$$\boxed{t = 0}$$

∴ since t-value not the same, lines do no meet.

∴ skew lines

d) $\begin{cases} x = 1+t \\ y = 2+t \\ z = -t \end{cases}$ ① $t \in \mathbb{R}$ $\qquad \begin{cases} x = 3-2u \\ y = 4-2u \\ z = -1+2u \end{cases}$ ② $u \in \mathbb{R}$

$\vec{m_2} = -2\vec{m_1}$

∴ parallel

check Pt $(3,4,-1)$ into ①.

$3 = 1+t$ $\quad \boxed{2 = t}$

$4 = 2+t$ $\quad \boxed{2 = t}$

$-1 = -t$ $\quad \boxed{1 = t}$

since t not the same, lines parallel but distinct.

e) $\dfrac{x-3}{4} = y-2 = z-2$ $\qquad \dfrac{x-2}{-3} = \dfrac{y+1}{2} = \dfrac{z-2}{-1}$ ②

$\vec{m_1} = (4,1,1)$ $\qquad \vec{m_2} = (-3,2,-1)$ $\qquad \vec{m_1} \not\parallel \vec{m_2}$

$\begin{cases} x = 3+4t \\ y = 2+t \\ z = 2+t \end{cases}$ ① $t \in \mathbb{R}$

∴ $(x,y,z) = (1,1,1)$

sub ① into ②

$$\frac{3+4t-2}{-3} = \frac{2+t+1}{2}$$

$$2(1+4t) = -3(3+t)$$

$$2+8t = -9-3t$$

$$11t = -11$$

$$\boxed{t = -1}$$ ← SAME →

$$\frac{2+t+1}{2} = \frac{2+t-2}{-1}$$

$$-1(3+t) = 2t$$

$$-3-t = 2t$$

$$\boxed{-1 = t}$$

5a) $\vec{r_1} = (1,-1,1) + t(3,2,1)$ $\quad \vec{m_1} = (3,2,1)$ $\quad \vec{m_1} \times \vec{m_2}$

$\vec{r_2} = (-2,-3,0) + u(1,2,3)$ $\quad \vec{m_2} = (1,2,3)$

$$\left.\begin{aligned} x &= 1+3t \\ y &= -1+2t \\ z &= 1+t \end{aligned}\right\} \text{①} \quad t \in \mathbb{R}$$

$$\frac{x+2}{1} = \frac{y+3}{2} = \frac{z}{3} \quad \text{②}$$

Sub ① into ②

$$\frac{1+3t+2}{1} = \frac{-1+2t+3}{2} \qquad \frac{-1+2t+3}{2} = \frac{1+t}{3}$$

$$2(3+3t) = 2+2t \qquad 3(2+2t) = 2(1+t)$$

$$6+6t = 2+2t \qquad 6+6t = 2+2t$$

$$4t = -4 \qquad 4t = -4$$

$$t = -1 \quad \longleftarrow \text{SAME} \longrightarrow \quad t = -1$$

$\therefore$ $x = 1+3(-1) = -2$

$y = -1+2(-1) = -3$

$z = 1+(-1) = \emptyset$.

$\therefore$ P.O.I. $(-2,-3,\emptyset)$

b) $\vec{m_2} = \vec{m_1} \times \vec{m_2}$

$= (4,-8,4)$

OR $= (1,-2,1)$

$$\begin{vmatrix} i & j & k & i & j \\ 2 & 1 & 3 & 2 & \\ 3 & 2 & 1 & 2 & 3 \end{vmatrix}$$

$= (4,-8,4)$

$\therefore$ $\vec{r} = (-2,-3,\emptyset) + s(1,-2,1)$, $s \in \mathbb{R}$

6) $\vec{r_1} = (4,7,-1) + t(4,8,-4)$, $t \in \mathbb{R}$

$\vec{r_2} = (1,5,4) + u(-1,2,3)$, $u \in \mathbb{R}$

$\vec{m_1} \cdot \vec{m_2}$

$= (4,8,-4) \cdot (-1,2,3)$

$= -4+16-12 = \emptyset$ ✓

$\because \vec{m_1} \perp \vec{m_2} \quad \therefore \vec{r_1} \perp \vec{r_2}$.

$\vec{r_1}$: $\left.\begin{aligned} x &= 4+4t \\ y &= 7+8t \\ z &= -1-4t \end{aligned}\right\}$ ① $t \in \mathbb{R}$

$\vec{r_2}$: $\frac{x-1}{-1} = \frac{y-5}{2} = \frac{z-4}{3}$ ②

Sub ① into ②

$$\frac{4+4t-1}{-1} = \frac{7+8t-5}{2} \qquad \frac{7+8t-5}{2} = \frac{-1-4t-4}{3}$$

$$2(3+4t) = -1(2+8t) \qquad 3(2+8t) = 2(-5-4t)$$

$$6+8t = -2-8t \qquad 6+24t = -10-8t$$

$$16t = -8 \qquad 32t = -16$$

$$t = -\frac{1}{2} \qquad t = -1$$

$\therefore$ P.O.I:

$x = 4+4(-\frac{1}{2}) = 2$

$y = 7+8(-\frac{1}{2}) = 3$

$z = -1-4(-\frac{1}{2}) = 1$

$\therefore (2,3,1)$

7) $\left.\begin{aligned} x &= 24+7t \\ y &= 4+t \\ z &= -20-5t \end{aligned}\right\}\ t\in\mathbb{R}$

x-intercept: set $y=0$, $z=0$

$x = 24+7t$ $\quad$ $0 = 4+t$ $\quad$ $0 = -20-5t$

$= 24+7(-4)$ $\quad$ $-4 = t$ $\quad$ $-4 = t$ $\quad$ $\therefore (-4,0,0)$

$x = -4$

y-intercept: set $x=0$, $z=0$

$0 = 24+7t$ $\quad$ $0 = -20-5t$ $\quad$ $\therefore$ since t-value not the same, no y-intercept

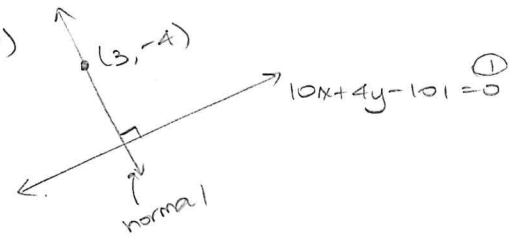
$-\frac{24}{7} = t$ $\quad$ $-4 = t$

z-intercept: set $x=0$, $y=0$

$0 = 24+7t$ $\quad$ $0 = 4+t$ $\quad$ $\therefore$ since t-value not the same, no z-intercept

$-\frac{24}{7} = t$ $\quad$ $-4 = t$

8)



$\vec{n} = (10,4)$

$\therefore \vec{r_2} = (3,-4) + t(10,4)$, $t\in\mathbb{R}$

$\left.\begin{aligned} x &= 3+10t \\ y &= -4+4t \end{aligned}\right\}\ t\in\mathbb{R}$ ②

sub ② into ①

$10(3+10t) + 4(-4+4t) - 101 = 0$

$30+100t-16+16t-101 = 0$

$116t = 87$

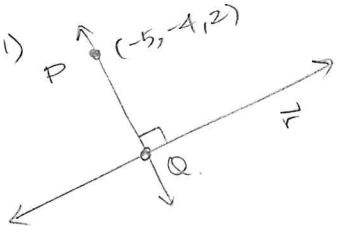
$t = \frac{3}{4}$

$\therefore x = 3+10\left(\frac{3}{4}\right) = \frac{21}{2}$

$y = -4+4\left(\frac{3}{4}\right) = -1$

$\therefore$ P.O.I $\left(\frac{21}{2}, -1\right)$

11)



$\vec{r} = (7,-13,8) + t(1,2,-2)$, $t \in \mathbb{R}$

$\vec{OQ} = (7+t, -13+2t, 8-2t)$

$\vec{PQ} = (7+t+5, -13+2t+4, 8-2t-2)$

$= (12+t, -9+2t, 6-2t)$

since $\vec{PQ} \perp \vec{r}$, $\vec{PQ} \perp \vec{m}$.

$\vec{PQ} \cdot \vec{m} = 0$

$(12+t, -9+2t, 6-2t) \cdot (1,2,-2) = 0$

$12+t-18+4t-12+4t = 0$

$9t = 18$

$t = 2$

$\therefore Q(9,-9,4)$

$\therefore \vec{r_2} = (-5,-4,2) + s\underbrace{(14,-5,2)}_{\vec{PQ}}$, $s \in \mathbb{R}$

12) $\vec{r} = (0,5,3) + t(1,-3,-2)$, $t \in \mathbb{R}$

$\left.\begin{array}{l} x = t \\ y = 5-3t \\ z = 3-2t \end{array}\right\}$ $t \in \mathbb{R}$ sub into sphere equation.

$x^2 + y^2 + z^2 = 6$

$(t)^2 + (5-3t)^2 + (3-2t)^2 = 6$

$t^2 + 9t^2 - 30t + 25 + 4t^2 - 12t + 9 = 6$

$14t^2 - 42t + 28 = 0$

$14(t^2 - 3t + 2) = 0$

$(t-2)(t-1) = 0$

$\therefore t = 2$ or $t = 1$

if $t = 1$

$x = 1$

$y = 5-3(1) = 2$

$z = 3-2(1) = 1$

$\therefore P_1(1,2,1)$

if $t = 2$

$x = 2$

$y = 5-3(2) = -1$

$z = 3-2(2) = -1$

$\therefore P_2(2,-1,-1)$

$\vec{P_1P_2} = (1,-3,-2)$

$|\vec{P_1P_2}| = \sqrt{1^2 + (-3)^2 + (-2)^2}$

$= \sqrt{14}$

diameter $= 2\sqrt{6}$

$= \sqrt{24} > \sqrt{14}$

$\therefore$ NO!

13) Determine P.O.I. of $\vec{r_1}$ & $\vec{r_2}$.

$\vec{r_1}$: $x = 2+t$, $y = -16+t$, $z = 19-4t$ } ① $t \in \mathbb{R}$

$\vec{r_2}$: $\frac{x-14}{-2} = \frac{y-19}{1} = \frac{z+2}{2}$ ②

$\frac{2+t-14}{-2} = \frac{-16+t-19}{1}$

$1(-12+t) = -2(-35+t)$

$-12+t = 70-2t$

$3t = 82$

$t = \frac{82}{3}$

$\frac{-16+t-19}{1} = \frac{19-4t+2}{2}$

$2(-35+t) = 1(21-4t)$

$-70+2t = 21-4t$

$6t = 91$

$t = \frac{91}{6}$

⟵ NOT SAME ⟶

∴ not a single point of intersection.

∴ line meets each line separately.

∴ $\vec{m_3}$ = origin to line 1

$= (2+t, -16+t, 19-4t)$

$\vec{m_4}$ = origin to line 2

$= (14-2u, 19+u, -2+2u)$

If origin to line 1 & line 2 makes a line,

then $\vec{m_3}$ // $\vec{m_4}$ or scalar multiples.

∴ $\vec{m_3} = k\vec{m_4}$, $k \in \mathbb{R}$

$(2+t, -16+t, 19-4t) = k(14-2u, 19+u, -2+2u)$

$2+t = k(14-2u) \Rightarrow 2 = -t+14k-2uk$ ③

$-16+t = k(19+u) \Rightarrow -16 = -t+19k+uk$. ④

$19-4t = k(-2+2u) \Rightarrow 19 = 4t-2k+2uk$. ⑤

③ $2 = -t + 14k - 2uk$

⑤ $19 = 4t - 2k + 2uk$

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④ $21 = 3t + 12k$ ⑥

③ $2 = -t + 14k - 2uk$

④ × 2 $-32 = -2t + 38k + 2uk$

$+ \quad -30 = -3t + 52k$ ⑦

⑥ $21 = 3t + 12k$

⑦ $-30 = -3t + 52k$

$+ \quad -9 = 64k$

$$\boxed{\frac{-9}{64} = k}$$

Sub $k = \frac{-9}{64}$ into ⑥

$$21 = 3t + 12\left(\frac{-9}{64}\right)_{16}$$

$$21 = 3t - \frac{27}{16}$$

$$336 = 48t - 27$$

$$363 = 48t$$

$$\frac{363}{48} = t$$

$$\boxed{\frac{121}{16} = t}$$

$\therefore \vec{m_3} = (2+t, -16+t, 19-4t)$

$= \left(2+\left(\frac{121}{16}\right), -16+\left(\frac{121}{16}\right), 19-4\left(\frac{121}{16}\right)\right)$

$= \left(\frac{153}{16}, \frac{-135}{16}, \frac{-180}{16}\right)$

or use $(153, -135, -180)$

OR $(17, -15, -20)$

$\therefore \vec{r} = (0,0,0) + s(17, -15, -20), \; s \in R$

$\vec{r} = s(17, -15, -20), \; s \in R$.